higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N4

19 November 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 Short V_{20}

$R_p = (6 \times 9)/(6 + 9)$
$= 3,6\ \Omega$

$R_S = 3,6 + 11$
$= 14,6\ \Omega$

$R_{p2} = (14,6 \times 3)/(14,6 + 3)$
$= 2,489\ \Omega$

$R_T = 2,489 + 4$
$= 6,489\ \Omega$

$I_{TA} = 28/6,489$
$= 4,315\ A$✓

$I_{3\Omega} = 14,6/(14,6 + 3) \times 4,315$
$= 3,579\ A$✓

Short V_{28}

$R_p = (4 \times 3)/(4 + 3)$
$= 1,714\ \Omega$

$R_S = 1,714 + 11$
$= 12,714\ \Omega$

$R_{p2} = (12,714 \times 9)/(12,714 + 9)$
$= 5,27\ \Omega$

$R_T = 6 + 5,27$
$= 11,27\ \Omega$

$I_{TB} = 20/11,27$
$= 1,775\ A$✓

$I_B = 9/(9 + 12,714) \times 1,736$
$= 0,736\ A$✓

$I_{B3\Omega} = 4/(4 + 3) \times 0,736$
$= 0,421\ A$✓

$I_{3\Omega} = 3,579 + 0,421$
$= 4\ A$✓ (6)

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1.2 $V_{th} = 3/(3 + 5) \times 32$
$= 12$ V✓✓

$R_{th} = (5 \times 3)/(5 + 3)$
$= 1,875\ \Omega$✓✓

$I = 12/(1,875 + 8)$
$= 1,215$ A✓✓ (6)
[12]

QUESTION 2

2.1 $BW = \Delta f$
$= 3\ 000 \times 10^3/200$
$= 15$ kHz✓

$F_L = 3\ 000 \times 10^3 + (15 \times 10^3/2)$✓
$= 2\ 992,5$ kHz✓

$F_H = 3\ 000 \times 10^3 + 15 \times 10^3/2$✓
$= 3\ 007,5$ kHz✓ (5)

2.2 $Z_1 = 75 + j5$
$= 75,166 \angle 3,814$✓

$Z_2 = 105 - j223$
$= 246,483 \angle -64,787$✓

$Z_3 = 0 + j128$
$= 128 \angle 90$✓

$Z_p = (75,166 \angle 3,814 \times 246,483 \angle -64,787)/(75 + 5 + 105 - j223)$
$= 18\ 527,141 \angle -60,973/(180 - j218)$
$= 18\ 527,141 \angle -60,973/282,708 \angle -50,454$
$= 65,535 \angle -10,519\ \Omega$✓

$Z_T = 64,434 - j11,964 + 0 + j128$
$= 132,726 \angle 60,957\ \Omega$✓✓

$I_T = 220 \angle 0/132,726 \angle 60,957$✓
$= 1,658 \angle -60,957$ A✓ (8)
[13]

QUESTION 3

3.1 $T = Rqi/K$
$= (7,233 \times 1,6 \times 10^{-19} \times 4 \times 10^{-3})/1,38 \times 10^{-23}$✓
$= 335,443$✓ (2)

3.2 $I = I_s(e^{qv/KT} - 1)$
$= 83 \times 10^{-9}(e^{23.50176} - 1)$✓
$= 1\ 335,869$ A✓ (2)

3.3 $V_S = V_T - V_z$
$= 15 - 8$
$= 7$ V

$I_z = P_z/V_z$
$= 5/8$
$= 0,625$ A✓

$I_s = I_z + I_L$
$= 0,625 + 0,16$
$= 0,785$ A✓

$R_s = V_s/I_s$
$= 7/0,785$
$= 8,917\ \Omega$✓✓

$R_L = V_L/I_L$
$= 7/0,16$
$= 43,75\ \Omega$✓✓ (6)
[10]

QUESTION 4

4.1 4.1.1 True
4.1.2 True
4.1.3 False
4.1.4 False
(4 × 1) (4)

4.2 4.2.1 $V_m = \sqrt{2} \times 30$
$= 42,426\ V$✓

$V_{dc} = V_m/(1 + 1/(2fCR_L))$
$= 42,426/(1 + 1/(2 \times 50 \times 620 \times 10^{-6} \times 100))$✓✓
$= 36,534\ V$✓ (4)

4.2.2 $R = 1/(2\sqrt{3}fCR_L)$
$= 1/(2 \times \sqrt{3} \times 50 \times 620 \times 10^{-6} \times 100)$✓
$= 0,093$✓

$V_{r(rms)} = 0,093 \times 36,534$✓
$= 3,402\ V$✓ (4)
[12]

QUESTION 5

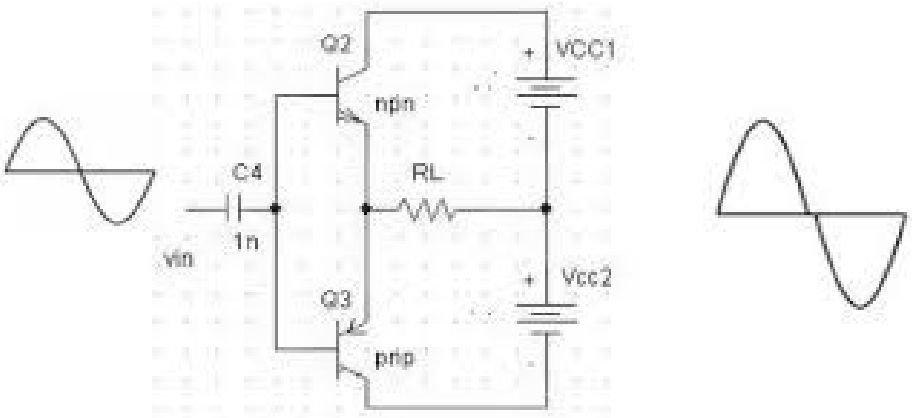
5.1 5.1.1 Linear region: When the base/emitter junction is forward-biased and the base collector is reverse-biased

5.1.2 Cut-off region: When both base/emitter and base/collector junctions are reverse-biased

5.1.3 Saturation: When both the base/emitter and the base/collector junctions are forward-biased

(3 × 2) (6)

5.2

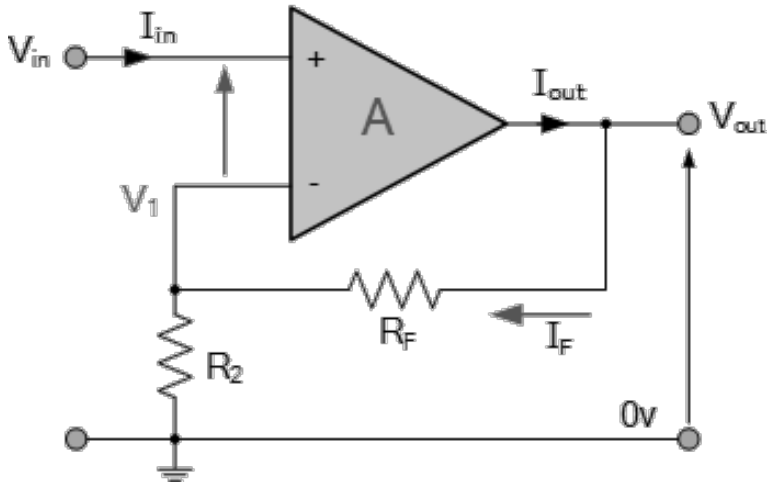


(4)

5.3 When the input and output signals are converted to currents, added and reconverted to a voltage and applied to the amplifier (2)
[12]

QUESTION 6

6.1

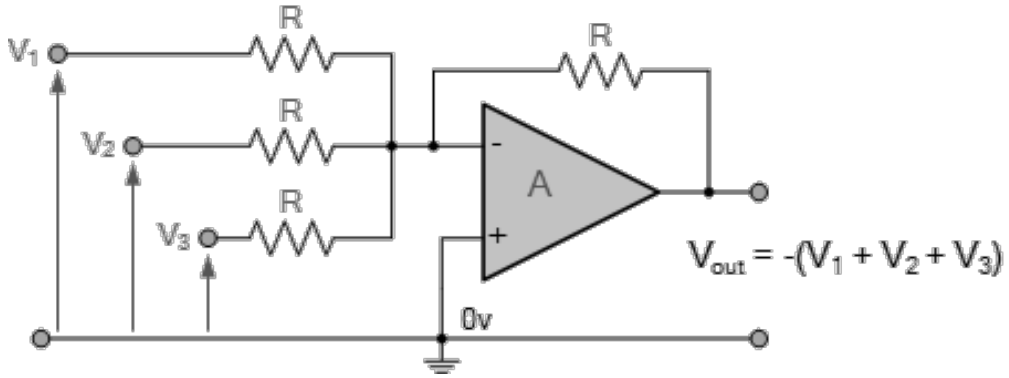


(4)

6.2 $A_v = V_o/V_{in}$
$= 8/0,35$
$= 22,857$✓

$R_{in} = R_f/A_V$
$= 30 \times 10^3/22,857$✓
$= 1,313\ k\Omega$✓ (3)

6.3



(4)
[11]

QUESTION 7

7.1 7.1.1 The SCR will turn on. The anode to cathode voltage will drop to the conduction value, resulting in a reduced gate current. (3)

7.1.2 The SCR will turn off. (2)

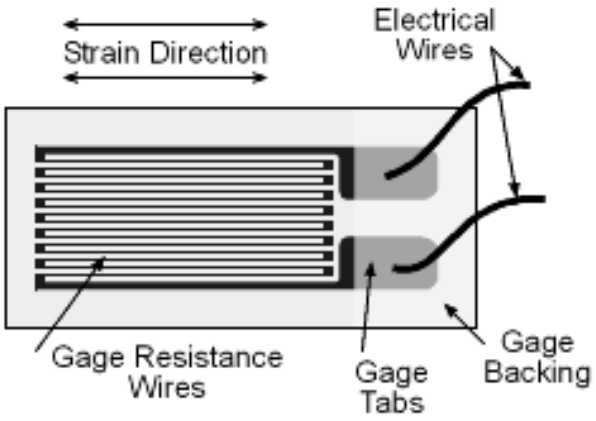
7.2 When the current falls below the holding current (2)

7.3 It is a two-terminal device which may be referred to as a bidirectional trigger diode or a diode AC switch. It is effectively two PNPN structure in parallel which allows it to be triggered in either direction. (3)
[10]

QUESTION 8

8.1 As pressure is applied to the Bourdon tube, the resulting force attempts to straighten the Bourdon tube. When the pressure is reduced, the Bourdon tube returns to its original shape. A potentiometer is connected to the rotating shaft of the Bourdon tube. When the Bourdon tube straightens, the shaft turns and the resistance of the potentiometer changes. (5)

8.2



(3)

8.3 Two or three terminals (2)
[10]

QUESTION 9

9.1 9.1.1 D
9.1.2 A
9.1.3 E
9.1.4 B
9.1.5 G
9.1.6 C

(6 × 1) (6)

9.2 9.2.1 $V_P = 2,9 \times 0,6$✓
$= 1,74$ V✓

9.2.2 $V_{rms} = 0,7071 \times 1,74$
$= 1,23$ V✓

$V_{ave} = 0,637 \times 1,74$
$= 1,108$ V✓

(2 × 2) (4)
[10]

TOTAL: 100